

Chapter 3 Waves & Light (up to 3.2.2)

30 focused Paper 2-style questions with answers for Cambridge O Level Physics 5054 revision

Question wording is taken from the supplied repository PDF: PHYSICS (2).pdf. The supplied file contains Paper 1 answer-key pages but does not appear to include Paper 2 mark schemes, so the answers below are concise model answers written to match expected Cambridge marking points.

Topic coverage	Question numbers
Wave motion, features, $v = f\lambda$	1-8, 11, 17-19
Transverse/longitudinal waves	5, 7-10
Reflection/refraction/diffraction of waves + ripple tank	12-19
Reflection of light + plane mirror	20-22
Refraction, refractive index, critical angle, TIR, optical fibres	23-30

3.1 General properties of waves

1. Source: 5054/21/O/N/14 Q4(a)

Ripple tanks are often used to illustrate wave motion.

Describe what is meant by wave motion in a ripple tank. [2]

Answer: Disturbance/waves travel across the water, transferring energy, while the water itself only oscillates up and down and is not carried along with the wave.

2. Source: 5054/21/O/N/14 Q4(b)(i)

When describing wave motion, state what is meant by frequency. [1]

Answer: The number of complete waves/oscillations/wavelengths passing a point each second.

3. Source: 5054/21/O/N/14 Q4(b)(ii)

When describing wave motion, state what is meant by wavelength. [2]

Answer: The distance between two successive points in phase, e.g. crest to next crest or compression to next compression.

4. Source: 5054/22/M/J/11 Q4(a)

Fig. 4.2 shows how the displacement of A varies with time.

State the amplitude of the vibrations of A as the wave passes. [1]

Answer: Amplitude = 2 mm. It is the maximum displacement from the mean/rest position.

5. Source: 5054/22/M/J/12 Q6(b)

Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.

On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X. [1]

Answer: Arrow vertically up or down: water molecules vibrate at right angles to the direction of travel.

6. Source: 5054/22/M/J/12 Q6(c)(i)

The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

Calculate the speed of the wave. [2]

Answer: $v = f\lambda = 2.0 \times 2.5 = 5.0$ cm/s, or 0.050 m/s.

7. Source: 5054/21/M/J/25 Q5(a)(i)

Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

Answer: C goes at a region where particles are closest together. R goes at a region where particles are furthest apart.

8. Source: 5054/21/M/J/25 Q5(a)(ii)

Describe the difference between a compression and a rarefaction. [1]

Answer: Compression: high pressure / particles close together. Rarefaction: low pressure / particles spread apart.

9. Source: 5054/21/O/N/18 Q9(a)

Some waves are longitudinal and some waves are transverse.
State how a longitudinal wave differs from a transverse wave. [1]

Answer: Longitudinal vibrations are parallel to the direction of wave travel/energy transfer; transverse vibrations are perpendicular to it.

10. Source: 5054/22/M/J/23 Q5(a)

Water waves are transverse waves.
Underline two other examples of transverse waves.
seismic P-waves seismic S-waves sound X-rays [1]

Answer: seismic S-waves and X-rays.

11. Source: 5054/21/M/J/25 Q5(b)(ii)

The frequency of the water wave is increased by moving the wooden bar up and down more quickly.
State what happens to the speed and what happens to the wavelength of the wave produced. [2]

Answer: Speed: unchanged (same depth). Wavelength: decreases.

12. Source: 5054/21/M/J/25 Q5(c)

Describe what is meant by the diffraction of a water wave. [1]

Answer: Spreading/bending of a wave as it passes through a gap or past an edge.

13. Source: 5054/22/M/J/23 Q5(b)(ii)

The wave diffracts at the right-hand edge of the glass block.
On Fig. 5.1 draw two crests after they pass the glass block to show the diffraction. [2]

Answer: Draw curved/spreading crests beyond the edge, centred roughly on the edge, with the same spacing as before if depth is unchanged.

14. Source: 5054/22/M/J/23 Q5(b)(iv)

Describe how a decrease in wavelength affects the diffraction. [1]

Answer: A smaller wavelength gives less diffraction / less spreading.

15. Source: 5054/21/M/J/23 Q7(b)

The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.
Complete Fig. 7.2 with at least three crests to show the new diffraction pattern. [2]

Answer: Draw wide semicircular crests spreading out from the gap; maximum diffraction occurs when gap size is smaller than/about equal to wavelength.

16. Source: 5054/21/O/N/14 Q4(c)

A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.
On Fig. 4.1, draw the reflected parts of these three wavefronts. [3]

Answer: Draw reflected wavefronts with the same spacing, reflected as if from a plane mirror: angle of reflection equals angle of incidence.

17. Source: 5054/21/O/N/12 Q5(b)

As the wave passes into the shallow water, state what happens to (i) the frequency, (ii) the speed. [2]

Answer: Frequency: unchanged. Speed: decreases in shallower water.

18. Source: 5054/02/O/N/05 Q3(a)

A student produces wavefronts in a ripple tank to demonstrate refraction.
State what is meant by a wavefront. [1]

Answer: A line joining points on a wave that are in phase, e.g. a crest line.

19. Source: 5054/02/O/N/05 Q3(b)

State what happens to each of the following quantities as the wavefronts change direction.
(i) wavelength (ii) speed (iii) frequency [3]

Answer: Entering shallower water: wavelength decreases, speed decreases, frequency remains unchanged.

3.2.1 Reflection of light

20. Source: 5054/22/O/N/12 Q10(b)

Describe an experiment to verify the law of reflection for light. You may include a diagram in your answer. [5]

Answer: Place a plane mirror on paper, draw its line and a normal. Shine a ray from a ray box at the mirror. Mark incident and reflected rays, remove mirror, draw rays, measure i and r with a protractor. Repeat for several angles; $i = r$.

21. Source: 5054/22/O/N/12 Q10(d)(ii)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.
State the distance between I and the mirror. [1]

Answer: 0.19 m. A plane mirror image is the same distance behind the mirror as the object is in front.

22. Source: 5054/22/O/N/12 Q10(d)(iii)

Suggest one advantage of placing a mirror behind the lamp in the room. [1]

Answer: More light is directed into the room / the room appears brighter / light that would go backwards is reflected forwards.

3.2.2 Refraction of light

23. Source: 5054/22/M/J/13 Q10(a)

On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence i and the angle of refraction r . [3]

Answer: Normal: line perpendicular to the surface at B. i : angle between incident ray and normal. r : angle between refracted ray inside glass and normal.

24. Source: 5054/22/M/J/13 Q10(b)

State, in terms of the properties of light waves, why the light refracts at B. [1]

Answer: The light changes speed when it enters glass from air, so its direction changes.

25. Source: 5054/21/O/N/23 Q4(b)(i)

State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block. [3]

Answer: Speed decreases; frequency remains unchanged; wavelength decreases.

26. Source: 5054/21/O/N/13 Q10(c)(ii)

The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this. [2]

Answer: Mark the path of the ray entering and leaving the prism/block with pencil dots; remove the glass; join the dots with straight lines; measure the angle to the normal with a protractor.

27. Source: 5054/22/M/J/10 Q5(a)(i)

The ray of light strikes the glass surface at M with an angle of incidence C equal to the critical angle of light in glass.

State what is meant by critical angle. [1]

Answer: The angle of incidence in the denser medium for which the angle of refraction is 90° / the refracted ray travels along the boundary.

28. Source: 5054/21/O/N/10 Q9(a)(ii)

Explain why the boy is unable to see the red light at Q when his eye is closer to the pool wall than P. [2]

Answer: The angle of incidence at the water-air surface is greater than the critical angle, so total internal reflection occurs and the ray does not emerge to reach the eye.

29. Source: 5054/22/O/N/12 Q10(c)(i)-(ii)

The angle between the ray and the side of the fibre is 7° .

(i) Determine the angle of incidence of the ray at P. [1]

(ii) State and explain what happens to the ray at P. [2]

Answer: (i) 83° . (ii) It is totally internally reflected because the angle of incidence is greater than the critical angle for the glass-air boundary.

30. Source: 5054/22/M/J/17 Q(b)(iv)

Optical fibres are used in telecommunications to send large amounts of data.

Pulses of light travel at almost the same speed along a fibre as electrical pulses travel along a metal wire.

State two advantages of using optical fibres rather than wires for transmitting data. [2]

Answer: Any two: carry more data/bandwidth; less signal loss/attenuation; less interference/crosstalk; more secure; lighter/smaller; cheaper in some contexts.

Tip: for diagram questions, redraw the figure from the original PDF source page, then compare your drawing with the written marking points here.